

ABHINAV PILLAI

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Education

University of Oklahoma

Bachelor of Science - Computer Science

Norman, OK

Aug 2022 - July 2026

Work Experience

University of Oklahoma

Undergraduate Research Assistant

Sep 2025 – May 2026

Norman, OK

- Engineered an **autonomous humanoid receptionist** on the Unitree G1 EDU Plus using ROS2, delivering real-time navigation, speaker tracking, and human-aware motion control.
- Built an on-device **AI pipeline** on NVIDIA Jetson Orin NX integrating **speech-to-text, LLM-driven dialogue, and retrieval-augmented generation (RAG)** for secure, offline knowledge delivery.
- Developed **facial recognition** and **VIP detection** using **depth sensing hardware and computer vision APIs**, enabling personalized interactions and adaptive response behaviors for high-profile users.

Software Studio @ University of Oklahoma

Startup Mentor and Teaching Assistant

Sep 2025 – May 2026

Norman, OK

- Mentored **40+ students** in building and launching production-ready software products, guiding architecture decisions, feature scoping, and MVP development.
- Led technical workshops on **user stories, feature matrices, prototyping, and system design** to help teams translate ideas into scalable software solutions.
- Advised founders on **product strategy, technical roadmapping, and go-to-market preparation**, supporting teams through live investor pitch cycles.

Vast Vertex Media

Full-stack Software Engineer Intern

Sep 2024 – Apr 2025

Norman, OK

- Built a production-ready **full-stack** mobile application using **React Native, TypeScript, PostgreSQL, Azure, Power BI, and REST APIs**, supporting production deployment and analytics dashboards.
- Designed and shipped various responsive, user-friendly websites for clients using **Next.js, Tailwind CSS, Figma, and REST APIs**

University of Oklahoma

Undergraduate Research Assistant

Aug 2023 – Dec 2024

Norman, OK

- Built ONEngineering, a full-stack **Next.js/Typescript based** platform connecting over **400+** active OU Graduate and alumni with industry professionals
- Engineered **Python and C++** data ingestion pipelines processing RSS feeds to group student and alumni records, cutting recommendation deployment time by **50%**.

UFLIP

Software Engineer Intern

Mar 2023 – Nov 2023

Oklahoma City, OK

- Built a responsive front-end using **CSS, SASS, Node.js, JS, HTML, and Figma**, serving **3000+** users
- Reduced website startup time by **60%** by replacing synchronous code with asynchronous functions and API calls

Projects

The Trail - Every Step Forward | [Source Code](#) | [Blog](#)

React Native | FastAPI | Python | MongoDB

- **Won best use of ElevenLabs at Hacklahoma 2026****
- Built a gamified productivity app converting focused screen time into game progress, integrating **HealthKit, Canvas LMS, and iOS Device Activity APIs** to track student engagement in real time
- Developed a **Gemini-powered AI insights engine** using a **Python/FastAPI** backend and **MongoDB**, with **ElevenLabs** voice synthesis delivering personalized reflection prompts to students

Technical Skills

Languages: Python, Java, C++, C#, TypeScript, JavaScript, SQL, NoSQL, HTML, CSS

AI/ML & Robotics: ROS2, Retrieval-Augmented Generation (RAG), LLM Integration, Computer Vision, Speech-to-Text

Frameworks & Tools: React, React Native, Next.js, Node.js, Docker, Git/GitHub, AWS EC2, Azure, Postman